FIRST SEMESTER 2021/2022 SESSION EXAMINATION
ECN 212: INTERMEDIATE STATISTICS 1
INSTRUCTION: Answer any 4 questions
Time: 2hrs 15mins

1a. Write short notes on the following terms: i) null set ii) finite set iii) subset iv) universal set v) set complement iv) Baye's law

1b. A coin is tossed 3 times. If X is the random variable of the occurrence of heads, obtain the sample space, the probability distribution of X and show that $\sum P_i = 1$.

2a. State 4 basic properties of a binomial probability distribution.

2b. A Social Scientist learns that 65% of the household in the city are without a refuse bin while 35% of the household in that city have refuse bins. If he were to visit 8 households in that city, what is the probability that he will find a refuse bin in:

i) 4 of the 8 household
ii) All of the 8 households
iii) 2 of the 8 households
iv) None of the 8 households

3a. let X be a uniform random variable. Outline its major characteristics

3b. suppose 4% of items manufactured by a certain factory are defective. Find the probability that in a sample of 100 items, there would be:

(ii) 4 defectives
(iii) At least 2 defectives

of defective items?

a. What is the relationship between normal, binomial and poisson distributions?

b. In a normal population, 15% values are less than 12 and 40% are above 16.2. Find the mean and standard deviation of this population.

a. Buses arrived a bus stop according to an exponential distribution with rate $\alpha = 4$ buses/hour.

(i) if you arrived at 8:00am to the bus stop, what is the expected time of the next bus?

(ii) assume you asked one of the people waiting for the bus about the arrival time of the last bus and he told you that the last bus left at 7:40am, what is the expected time of the next bus?

. It is known that 55% of civil servants gain promotion in the first three years of service. Using the normal approximation to the binomial distribution, calculate the probability that 50% of a group of 230 civil servants will gain promotion in their first three years.

UNIVERSITY OF UYO
DEPARTMENT OF ECONOMICS
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Time: 2hrs. 15mins

1a. Two dice are thrown. What is the mathematical expectation and variance of X?
1b. Discuss the addition and complementary laws of probability.
1c. In a gathering of technicians, 80 of them are electricians, 200 auto-mechanics, while 65 are plumbers. Calculate the probability of getting an electrician or a motor mechanic or a plumber in a single draw.

2a. State 4 basic properties of a binomial probability distribution.
2b. Suppose that the probability of parents having a child with blond hair is ¼. If there are 6 children in the family, what is the probability that half of them will have blond hair?
2c. If the probability of hitting a target on a single shot is 0.3, what is the probability that in 4 shots the target will be hit at least 3 times?

3a. With appropriate illustrations, distinguish among normal, uniform, and exponential distributions.
3b. Suppose 4% of items manufactured by a certain factory are defective. Find the probability that in a sample of 100 items, there would be:
(v) 4 defectives
(vi) At least 2 defectives
(vii) What are the expected value, standard deviation and variance of the number of defective items?

4a. What is the probability that a standard normal random variable takes a value between (i) -2 and +2 (ii) 0.3 and 3.2
4b. In a normal population, 15% values are less than 12 and 40% are above 16.2. Find the mean and standard deviation of this population.

5a. Buses arrive to a bus stop according to an exponential distribution with rate $\alpha = 4$ buses/hour.
(i) if you arrived at 8:00am to the bus stop, what is the expected time of the next bus?
(ii) assume you asked one of the people waiting for the bus about the arrival time of the last bus and he told you that the last bus left at 7:40am, what is the expected time of the next bus?
5b. Write short notes on the following: (i) an event (ii) Mutually exclusive event (iii) Independent event (iv) Dependent event (v) Venn diagram

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